OIL FUTURES

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WHAT IS OIL FUTURES

DERIVATIVES MARKET STRUCTURE | INSTRUMENT DEFINITION

Oil Future: A standardized exchange-traded contract to buy or sell crude oil at a specified price for a future delivery month.

Crude Oil (WTI/Brent)

Exchange Traded

Marked to Market Daily

TRADING VENUES

NYMEX (CME) — CL

Physical Delivery

Traders must take/deliver oil. Led to **negative prices** in 2020 due to storage constraints.

ICE (Intercont.) — B

Cash Settled

Optional delivery. Favored by **financial speculators** as it carries no logistics pressure.

CONTRACT SPECIFICATIONS

SPEC	WTI (NYMEX)	BRENT (ICE)
Size	1,000 bbl	1,000 bbl
Tick	$0.01 ($10)	$0.01 ($10)
Hours	Sun-Fri 5pm-4pm CT	~24h Electronic
Term	10yr + 2mo	96 months
Settle	**Physical**	**Cash/EFP**

MARKET CHARACTERISTICS

High Liquidity

Most actively traded commodity derivatives globally, ensuring deep markets.

Global Benchmark

Future prices serve as primary global reference benchmarks for physical crude.

24-Hour Trading

Electronic trading occurs nearly 24/7, allowing immediate reaction to events.

CRUDE OIL FUTURE PAYOFF ANALYSIS (LINEAR)

DERIVATIVES PRICING MECHANICS | LONG POSITION

Linear Payoff Structure

Unlike options with non-linear convexity, futures contracts have a symmetrical **linear payoff**. P&L changes 1:1 with underlying price movement.

Key Formula

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Payoff ($) =
  (ST - F0) × Size
ST: Spot@Expiry F0: Entry
```

Scenario Analysis

Long 1 Contract (1,000 bbl) @ $80.00

Bull Case

Spot ST = $90 — **+$10,000**

($90 - $80) × 1,000

Bear Case

Spot ST = $70 — **-$10,000**

($70 - $80) × 1,000

Long Position Payoff

Buying a Brent Crude Future

WHY OIL FUTURES ARE USEFUL

ECONOMIC FUNCTIONS | MARKET UTILITY

1. Hedging

RISK MANAGEMENT

A critical tool for market participants to manage exposure to price volatility, ensuring operational stability.

Real-World Examples

- **Airlines**
  Hedging jet fuel costs to protect profit margins against sudden price spikes.
- **Oil Producers**
  Locking in future selling prices to guarantee revenue and budget stability.

2. Price Discovery

MARKET SIGNALS

The central mechanism for global price formation, reflecting collective market expectations.

Key Mechanisms

- **Global Benchmarks**
  Brent and WTI futures serve as the primary reference prices for physical crude trading worldwide.
- **Future Expectations**
  Futures prices aggregate global information to reflect consensus on future supply/demand.

3. Liquidity

& SPECULATION

Provides essential market depth and facilitates efficient risk transfer between participants.

Market Functions

- **Risk Absorption**
  Speculators take the other side of hedgers' trades, absorbing price risk that hedgers want to shed.
- **Market Depth**
  High trading volumes ensure participants can enter/exit positions easily without moving prices significantly.

Major Market Participants

1. Commercial Hedgers

Oil Producers

Short futures to lock in selling price

ExxonMobil

Refiners

Manage crack spread: buy crude, sell product futures

Marathon

Airlines

Hedge jet fuel with oil/diesel futures

Delta

2. Commodity Trading Houses

Arbitrage & inventory management

Calendar spreads, geographic arbitrage (WTI vs Brent), quality arbitrage (light/sweet vs heavy/sour)

Vitol

3. Financial Investors

Hedge Funds	Directional & term-structure trades	*Citadel*
CTA / Systematic	Trend following & momentum	*Winton*
Index Funds	Passive long-only exposure	*BlackRock*
Market Makers	Provide liquidity & tight spreads	*Citadel*

01 Price Volatility

Geopolitics, OPEC production decisions, demand shocks, and USD fluctuations drive sudden price swings. These factors can rapidly alter market fundamentals and create significant P&L volatility.

Geopolitics | OPEC | USD Swings

02 Liquidity Risk

Front-month contracts are deeply liquid (1M+ contracts/day), but long-dated contracts trade much thinner — creating execution risk and potential slippage when entering or exiting large positions.

Front-Month | Thin Markets | Slippage

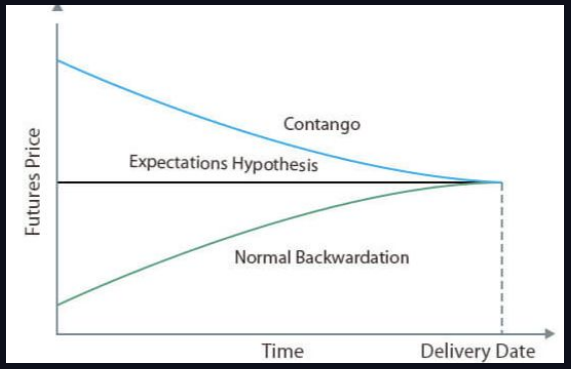
03 Basis Risk

Spot vs. futures price mismatch can erode hedging effectiveness. WTI-Brent spread also fluctuates significantly by region and logistics constraints, impacting cross-market hedging strategies.

WTI-Brent | Spot/Future | Logistics

04 Roll Risk

In contango markets, rolling long positions forward is costly (negative roll yield). In backwardation, rollers may earn a positive yield. Critical for ETFs and index funds managing long-term exposure.



Contango | Backwardation

MARKET MECHANICS

WHY OIL REACTS STRONGLY TO GEOPOLITICS | STRUCTURAL ANALYSIS

01 Supply Concentration

Oil reserves are unevenly distributed, creating critical dependencies on specific regions.

Key Chokepoints

- Strait of Hormuz
- Strait of Malacca

02 Supply Uncertainty

Conflicts, sanctions, and disruptions create immediate threats to physical availability.

Active Conflict

Direct damage to infrastructure (refineries, pipelines) reduces immediate output.

Sanctions Regimes

Artificial barriers remove barrels from the global market, tightening balances.

Shipping Disruptions

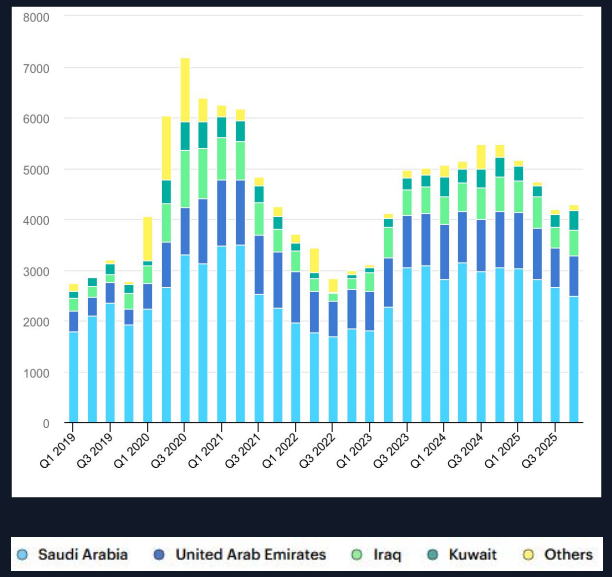
Insurance spikes and route diversions delay cargo arrival times.

IMPACT VELOCITY

03 Erasing the Buffer

OPEC ability to pump extra oil is at its lowest point in year

OPEC effective spare capacity, 2019-2025



U.S. OIL FUTURES PRICE (WTI)

APR 2026 CONTRACT | NOV 2025 - MAR 2026

EVENT TIMELINE // 2024-2026 CRITICAL PATH

2024 - 2025

STRUCTURAL **Eurasian Trade Fragmentation**

Event / Impact

Deepening secondary sanctions on Russia; trade flows re-routed.

Correlation

Set structural risks for future shocks

JAN 2026

SUPPLY SHOCK **U.S. Intervention in Venezuela**

Event / Impact

Regime change & naval blockades. Exhausted global spare capacity, removing the market's safety buffer.

Correlation

Removed market buffer, zero tolerance for subsequent shocks.

FEB 28, 2026

CONFLICT **Outbreak of Iran Conflict**

Event / Impact

U.S./Israel strikes on Iranian energy

Correlation

Immediate Spark.

MAR 02-05

BLOCKADE **Hormuz Strait Blockade**

Event / Impact

Effective closure of vital oil chokepoint (20% supply). Shifted market from "price risk" to **"logistical paralysis."**

Correlation

Amplified local disruption into a global hard deficit.

MAR 06-09

FINANCIAL **The "Gamma Magnet" Effect**

Event / Impact

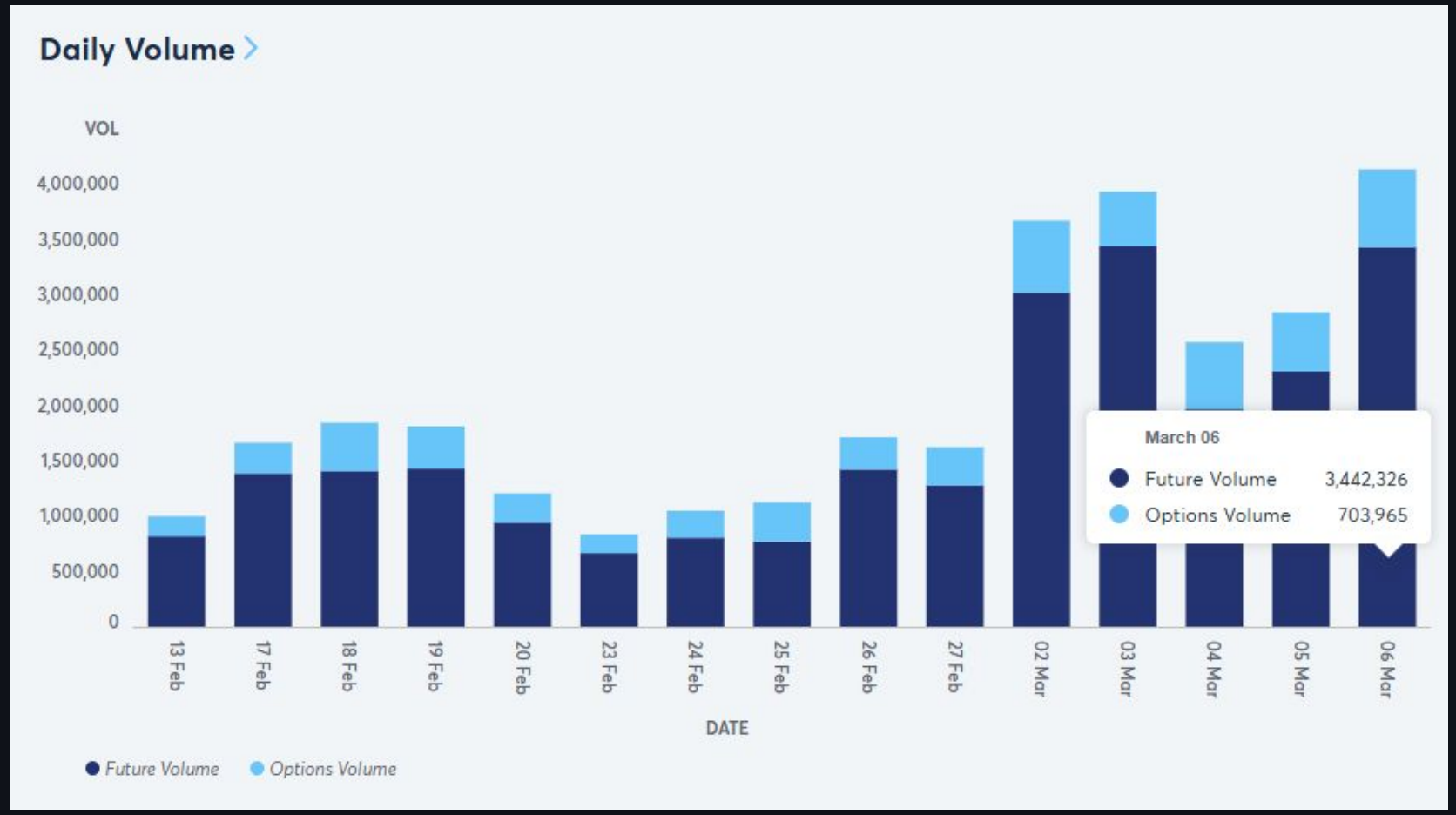
systematic deleveraging & dealer `Gamma Hedging`. Supply shock -> liquidity crisis.

Correlation

Financial amplifier: Supply shock turned into crisis.

RISK AMPLIFICATION

DERIVATIVES MARKET DYNAMICS | SYSTEMIC FEEDBACK LOOPS

Dealer Gamma Exposure

When market makers (dealers) are **Short Gamma**, they must hedge in the direction of the market—buying as prices rise and selling as prices fall. This hedging activity amplifies volatility, creating self-reinforcing price loops known as a "Gamma Squeeze" or "Unwinding".